Relaxation and Odd-Even Effect in Fermi Liquids

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Outline

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- k-space
- Fermi surface

2 Recovering equations

- Linearization of the collision integral
- Eigenvalue problem

3 Odd-Even effect

1 Introduction

1.1 k-space

k-space

- Analogous to momentum space
- Each point $\mathbf{k}$ in k-space corresponds to a wave $\mathrm{e}^{\mathrm{i}\mathbf{k}}$
- **Fourier transform** converts between k-space to real space

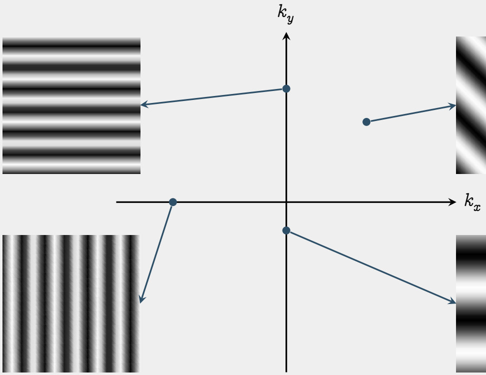

Figure 1: REPRESENTATIVE WAVES OF k-SPACE

- Distance from origin $\propto$ Wave frequency
- Direction on k-space $\implies$ Wave direction

1.2 Fermi surface

2 Recovering equations

2.1 Linearization of the collision integral

Ideas

- Consider **only perturbation near Fermi's equilibrium**

$$f(t,\mathbf{p}) = f_0(\mathbf{p}) + \delta f(t,\mathbf{p})$$

- Transform the integrals into a linear operator

Preparation

Fermi equilibrium: $f_0(\mathbf{p}) = \dfrac{1}{\exp[\beta(\varepsilon(\mathbf{p}) - \mu)] + 1}$

$$\frac{\partial f_0}{\partial \varepsilon} = -\frac{1}{T} f_0(1 - f_0) \quad (1)$$

$$\frac{\partial f_0}{\partial \mu} = \frac{1}{T} f_0(1 - f_0) \quad (2)$$

$$\frac{\partial f_0}{\partial \varepsilon} = -\frac{\partial f_0}{\partial \mu} \quad (3)$$

$$f(t,\mathbf{p}) = f_0(\mathbf{p}) + \delta f(t,\mathbf{p}) \qquad \text{Perturbation} \tag{4}$$

$$\approx f_0(\mathbf{p}) + \frac{\partial f_0}{\partial \mu}\,\delta\mu \qquad \text{Taylor series. Fix } \varepsilon \tag{5}$$

$$\therefore \delta f(t,\mathbf{p}) = \frac{\partial f_0}{\partial \mu}\,\delta\mu \tag{6}$$

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$$\therefore \delta f(t,\mathbf{p}) = \frac{\partial f_0}{\partial \mu}\,\delta\mu \quad (6)$$

> **Perturbation of interest:** $\delta\mu = T\psi$ (temperature dependent change in chemical potential)

$$\delta f(t,\mathbf{p}) = \frac{\partial f_0}{\partial \mu} T\psi \tag{7}$$

$$= \left(-\frac{1}{T} f_0(1-f_0)\right) \times T\psi \tag{8}$$

$$= f_0(1-f_0)\psi \tag{9}$$

Collision integral

$$\hat{\mathcal{J}}[f] = -\int \dots \int \frac{1}{(2\pi)^6} \, \mathrm{d}\mathbf{p}_2 \, \mathrm{d}\mathbf{p}'_1 \, \mathrm{d}\mathbf{p}'_2 W(\mathbf{p}'_1, \mathbf{p}'_2 | \mathbf{p}_1, \mathbf{p}_2) \mathcal{B}, \tag{10}$$

$$\mathcal{B} = f_1 f_2 (1 - f_{1'})(1 - f_{2'}) - f_{1'} f_{2'} (1 - f_1)(1 - f_2), \tag{11}$$

$$f_i = f(\mathbf{p}_i), \quad f_{0,i} = f_0(\mathbf{p}_i). \tag{12}$$

Linearization

Taylor expansion of $\mathcal{B}$ at f_0:

$$\mathcal{B} = \mathcal{B}_{f_0} + \mathcal{B}_{\text{lin}} + \mathcal{B}_{\text{non-lin}} \tag{13}$$

The linear part is:

$$\mathcal{B}_{\text{lin}} = \left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 + \left(\frac{\partial \mathcal{B}}{\partial f_2}\right)_{f_0} \delta f_2 + \left(\frac{\partial \mathcal{B}}{\partial f_1'}\right)_{f_0} \delta f_{1'} + \left(\frac{\partial \mathcal{B}}{\partial f_{2'}}\right)_{f_0} \delta f_{2'} \tag{14}$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} = f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'}) + f_{0,1'}f_{0,2'}(1 - f_{0,2}), \tag{15}$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_2}\right)_{f_0} = f_{0,1}(1 - f_{0,1'})(1 - f_{0,2'}) + f_{0,1}f_{0,2}(1 - f_{0,1}), \tag{16}$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_{1'}}\right)_{f_0} = -f_{0,1}f_{0,2}(1 - f_{0,2'}) - f_{0,2'}(1 - f_{0,1})(1 - f_{0,2}), \tag{17}$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_{2'}}\right)_{f_0} = -f_{0,1}f_{0,2}(1 - f_{0,1'}) - f_{0,1'}(1 - f_{0,1})(1 - f_{0,2}). \tag{18}$$

Substitute $\delta f_i = f_{0,i}(1 - f_{0,i})\psi_i$. E.g.,

$$\left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 = \Big[f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'}) + f_{0,1'}f_{0,2'}(1 - f_{0,2})\Big] \times f_{0,1}(1 - f_{0,1})\psi_1 \tag{19}$$

$$= \Big[f_{0,1}f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'})(1 - f_{0,1}) + f_{0,1}f_{0,1'}f_{0,2'}(1 - f_{0,1})(1 - f_{0,2})\Big]\psi_1$$

$$= \Big(f_{0,1}f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'})\Big)\Big[(1 - f_{0,1}) + f_{0,1}\Big]\psi_1 \tag{20}$$

$$= \Big(f_{0,1}f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'})\Big)\psi_1 \tag{21}$$

Define $F_{121'2'} = f_{0,1}f_{0,2}(1 - f_{0,1'})(1 - f_{0,2'}) = f_{0,1'}f_{0,2'}(1 - f_{0,1})(1 - f_{0,2})$. So that

$$\left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 = F_{121'2'}\psi_1 \tag{22}$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_i}\right)_{f_0} \delta f_i = F_{121'2'} \psi_i \tag{23}$$

Hence,

$$\mathcal{B}_{\text{lin}} = \left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 + \left(\frac{\partial \mathcal{B}}{\partial f_2}\right)_{f_0} \delta f_2 + \left(\frac{\partial \mathcal{B}}{\partial f_1'}\right)_{f_0} \delta f_{1'} + \left(\frac{\partial \mathcal{B}}{\partial f_{2'}}\right)_{f_0} \delta f_{2'} \tag{24}$$

$$= F_{121'2'} \left[\psi_1 + \psi_2 - \psi_{1'} - \psi_{2'}\right] \tag{25}$$

$$= F_{121'2'} \left[\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2')\right] \tag{26}$$

$$\hat{\mathcal{J}}[f] = -\frac{1}{(2\pi)^6}\int \dots \int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{p}_1\,\mathrm{d}\mathbf{p}_2' W(\mathbf{p}_1',\mathbf{p}_2'|\mathbf{p}_1,\mathbf{p}_2)\mathcal{B} \tag{27}$$

$$= -\frac{1}{(2\pi)^6}\int \dots \int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{p}_1\,\mathrm{d}\mathbf{p}_2' W(\mathbf{p}_1',\mathbf{p}_2'|\mathbf{p}_1,\mathbf{p}_2)(\mathcal{B}_{f_0} + \mathcal{B}_{\text{lin}} + \mathcal{B}_{\text{non-lin}}) \tag{28}$$

- Collision integral vanishes at equilibrium: $\hat{\mathcal{J}}[f_0] = 0$
- Throw out $\mathcal{B}_{\text{non-lin}}$

$$\hat{\mathcal{J}}[f] = -\frac{1}{(2\pi^6)}\int \dots \int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{p}_1\,\mathrm{d}\mathbf{p}_2'$$
$$\times W(\mathbf{p}_1',\mathbf{p}_2'|\mathbf{p}_1,\mathbf{p}_2)F_{121'2'}[\psi(\mathbf{p}_1)+\psi(\mathbf{p}_2)-\psi(\mathbf{p}_1')-\psi(\mathbf{p}_2')] \tag{29}$$

Expression of $F_{121'2'}$

- For convenience, write $F_{121'2'}$ as **total momentum** (P) and **center of momentum** (Q)

$$\mathrm{P} = \mathbf{p}_1 + \mathbf{p}_2, \quad \mathrm{Q} = \frac{1}{2}\mathbf{p}_1 - \frac{1}{2}\mathbf{p}_2. \tag{30}$$

$$\mathbf{p}_1 = \frac{1}{2}\mathbf{P} - \mathbf{Q}, \quad (31)$$

$$\mathbf{p}_2 = \frac{1}{2}\mathbf{P} + \mathbf{Q}, \quad (32)$$

$$\mathbf{P}\cdot\mathbf{Q} = \frac{1}{2}\left(p_1^2 - p_2^2\right). \quad (33)$$

$$\varepsilon(\mathbf{p}_1) = \frac{\hbar^2 p_1^2}{2m^*} = \frac{\hbar^2}{2m^*}\left(\frac{1}{4}P^2 + Q^2 - \mathbf{P}\cdot\mathbf{Q}\right), \quad (34)$$

$$\varepsilon(\mathbf{p}_2) = \frac{\hbar^2 p_2^2}{2m^*} = \frac{\hbar^2}{2m^*}\left(\frac{1}{4}P^2 + Q^2 + \mathbf{P}\cdot\mathbf{Q}\right). \quad (35)$$

Define[1]

$$\xi = \frac{\hbar^2\beta}{2m^*}, \quad \text{and} \quad X = \beta\left(\frac{\hbar^2 P^2}{8m^*} + \frac{\hbar^2 Q^2}{2m^*} - \mu\right) \tag{36}$$

Then,

$$\beta\big(\varepsilon(\mathbf{p}_1) - \mu\big) = \beta\left(\frac{\hbar^2}{8m^*}P^2 + \frac{\hbar^2}{2m^*}Q^2 - \mu\right) - \frac{\hbar^2\beta}{2m^*}\mathbf{P}\cdot\mathbf{Q} \tag{37}$$

$$= X - \xi\mathbf{P}\cdot\mathbf{Q}. \tag{38}$$

Similarly,

$$\beta\big(\varepsilon(\mathbf{p}_2) - \mu\big) = X + \xi\mathbf{P}\cdot\mathbf{Q}. \tag{39}$$

[1] X can possibly be interpretted as: *average deviation from background chemical potential*

Hence,

$$f_0(\mathbf{p}_1) = \frac{1}{\exp[\beta(\varepsilon(\mathbf{p}_1) - \mu)]} = \frac{1}{\exp[X - \xi \mathbf{P} \cdot \mathbf{Q}]}. \tag{40}$$

$$f_0(\mathbf{p}_2) = \frac{1}{\exp[\beta(\varepsilon(\mathbf{p}_1) - \mu)]} = \frac{1}{\exp[X + \xi \mathbf{P} \cdot \mathbf{Q}]} \tag{41}$$

Recall that $F_{121'2'} = f_0(\mathbf{p}_1)f_0(\mathbf{p}_2)(1 - f_0(\mathbf{p}_1'))(1 - f_0(\mathbf{p}_2'))$

$$f_0(\mathbf{p}_1)f_0(\mathbf{p}_2) = \frac{1}{\exp[X - \xi \mathrm{P} \cdot \mathrm{Q}] \exp[X + \xi \mathrm{P} \cdot \mathrm{Q}]} \tag{42}$$

$$= \frac{1}{\exp[2X] + \exp[X](\exp[\xi \mathrm{P} \cdot \mathrm{Q}] + \exp[-\xi \mathrm{P} \cdot \mathrm{Q}]) + 1} \tag{43}$$

$$= \frac{1}{\exp[2X] + 2\exp[X]\cosh(\xi \mathrm{P} \cdot \mathrm{Q}) + 1} \tag{44}$$

$$= \frac{1}{2\left(\cosh X + \cosh(\xi \mathrm{P} \cdot \mathrm{Q})\right)} \tag{45}$$

By symmetry,

$$(1 - f_0(\mathbf{p}_1'))(1 - f_0(\mathbf{p}_2')) = \frac{1}{2\left(\cosh X + \cosh(\xi \mathrm{P}' \cdot \mathrm{Q}')\right)} \tag{46}$$

$$F_{121'2'} = \frac{1}{4[\cosh X + \cosh(\xi \mathrm{P} \cdot \mathrm{Q})][\cosh X + \cosh(\xi \mathrm{P} \cdot \mathrm{Q})]} \tag{47}$$

Expression of W – the collision rate

$$W(\mathbf{p}_1', \mathbf{p}_2' | \mathbf{p}_1, \mathbf{p}_2) = \frac{2\pi}{\hbar} |V|^2 \times (2\pi)^2 \delta(\mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}_1' - \mathbf{p}_2') \delta\left(\varepsilon(\mathbf{p}_1) + \varepsilon(\mathbf{p}_2) - \varepsilon(\mathbf{p}_1') - \varepsilon(\mathbf{p}_2')\right)$$

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2.2 Eigenvalue problem

Reducing the Boltzmann equation

$$\left(\frac{\partial}{\partial t} + \mathbf{v} \cdot \frac{\partial}{\partial \mathbf{r}} + \mathbf{F} \cdot \frac{\partial}{\partial \hbar \mathbf{p}}\right) f(t, \mathbf{r}, \mathbf{p}) = \hat{\mathcal{J}}[f(t, \mathbf{r}, \mathbf{p})] \quad (49)$$

- No external force: $\mathbf{F} = \mathbf{0}$,
- No thermal gradient: $\frac{\partial f}{\partial \mathbf{r}} = 0$.

The equation reduces to

$$\frac{\partial f}{\partial t} = \hat{\mathcal{J}}[f] \quad (50)$$

Proposed ansatz $\psi(t, \mathbf{p}) = \mathrm{e}^{-\gamma t}\phi(\mathbf{p})$ (Chemical potential decays with time)

3 Odd-Even effect

Evaluation of $W(\mathbf{p}_1, \mathbf{p}_2 | \mathbf{p}_1', \mathbf{p}_2')$

$$W(\mathbf{p}_1, \mathbf{p}_2 | \mathbf{p}_1, \mathbf{p}_2) = \frac{2\pi}{\hbar} |V|^2 \times (2\pi)^2 \delta(\mathbf{p}) \tag{51}$$